

# Helixo AI: An Intelligent Edge-Computing Framework for Real-Time Helmet Compliance and Proactive Ignition Interlock Control

## 1. PROPOSED SOLUTION

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- **Proactive Enforcement:** An AI-based smart camera system integrated directly into the two-wheeler's instrument cluster.
- **Ignition Interlock:** The system physically restricts the vehicle's engine from starting unless it positively detects the rider wearing a safety helmet.
- **Continuous Monitoring:** Utilizes a 50-frame sliding window algorithm to monitor compliance continuously, initiating a safe, progressive engine cut-off protocol if the helmet is removed during the ride.

## 2. INNOVATION / NOVELTY

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- **Preventive vs Reactive:** Shifts traffic enforcement from a reactive penalty-based model (e.g., traffic cameras issuing fines post-facto) to a preventive model that stops the violation before it happens.
- **100% Offline Edge AI:** Computations run entirely locally on a microchip without the need for cloud infrastructure, internet connectivity, or smartphones.
- **Universal Compatibility:** Unlike RFID or Bluetooth-based smart helmets that require expensive custom gear, Helixo AI uses computer vision to detect *any* standard off-the-shelf helmet.

## 3. TECHNOLOGY STACK

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- **Artificial Intelligence & ML:** YOLOv8-nano (Ultralytics), Computer Vision (OpenCV), Python.
- **Edge Hardware:** ESP32-S3 (N16R8) or Kendryte K210 AI microcontrollers.
- **Sensors & Electronics:** OV2640/OV5640 Camera Module, 850nm IR LEDs (for night vision), 5V Opto-isolated Relay, LM2596 Buck Converter.

- **Web & Deployment:** ONNX Runtime Web, TensorFlow Lite INT8 Quantization, HTML/CSS/JS.

## 4. REAL-WORLD IMPACT

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- **Fatality Reduction:** Drastically reduces the severity of head injuries and fatalities in two-wheeler accidents by ensuring mandatory helmet compliance.
- **Automated Discipline:** Enforces strict traffic rules without requiring constant physical police presence, especially in rural or unmonitored areas.
- **Economic Relief:** Lowers the immense burden on public healthcare systems and families by preventing preventable traumatic brain injuries.

## 5. SDGS ADDRESSED

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- **SDG 3 (Good Health & Well-being):** Directly addresses Target 3.6 to halve the number of global deaths and injuries from road traffic accidents.
- **SDG 9 (Industry, Innovation & Infrastructure):** Contributes by innovating a cost-effective, indigenous TinyML hardware solution.
- **SDG 11 (Sustainable Cities and Communities):** Promotes safe, accessible, and sustainable transport systems for all citizens.

## 6. PROTOTYPE STATUS

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- **Current Stage:** Working Prototype / Proof of Concept (PoC) Ready (TRL-4/5).
- **AI Model:** Custom YOLOv8-nano model trained on 6,000+ local images for 115 epochs, achieving 99.87% precision and 97.50% mAP@50.
- **Demonstration:** A functional web-based simulation is live and accessible at <https://imaginative-kitsune-6e83b9.netlify.app/>. Hardware circuit logic is fully mapped and tested.